

Cherry-picking by trialists and meta-analysts can drive conclusions about intervention efficacy

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Abstract

Objectives: The objective of this study was to determine whether disagreements among multiple data sources affect systematic reviews of randomized clinical trials (RCTs).

Study Design and Setting: Eligible RCTs examined gabapentin for neuropathic pain and quetiapine for bipolar depression, reported in public (e.g., journal articles) and nonpublic sources (clinical study reports [CSRs] and individual participant data [IPD]).

Ethics approval: The study received an exemption from the Johns Hopkins Bloomberg School of Public Health Institutional Review Board (IRB No: 00006324).

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Conflict of interest: Up to 2008, K.D. served as an unpaid expert witness for the plaintiffs' lawyers in litigation against Pfizer that provided several gabapentin documents used for this study. S.V. was paid by the plaintiffs' attorneys for research assistance provided to K.D. for her work as an expert witness. Greene LLP, a law firm that has represented various clients in litigation against Pfizer, including the case in which K.D. served as an expert witness, provided a fund for scholarly research on reporting biases that partially covered salaries of some investigators performing analysis and interpretation of data. T.C. is President of The TMJ Association, Ltd., which received funding from Pfizer from 2009 to 2011 (for the Campaign to End Chronic Pain in Women) and from Purdue Pharma, L.P.

in 2008, 2010, and 2012 to 2016 (to support the TMJ Association science meeting and develop educational materials based on scientific research advances). P.D. was a corecipient of a UK National Institute for Health Research grant (HTA – 10/80/01 Update and amalgamation of two Cochrane Reviews: neuraminidase inhibitors for preventing and treating influenza in healthy adults and children—<http://www.nets.nihr.ac.uk/projects/hta/108001>). P.D. received €1,500 from the European Respiratory Society in support of his travel to the society's September 2012 annual congress in Vienna, where he gave an invited talk on oseltamivir. P.D. gratefully acknowledges the American Association of Colleges of Pharmacy for its funding support (\$11,000) for a study to analyze written medical information regarding the possible harms of statins. AACP had no involvement in the design and conduct of the study; collection, management, analysis, and interpretation of the data; and preparation, review, or approval of this manuscript. P.D. is also an associate editor of The BMJ and an unpaid member of the IMEDS steering committee at the Reagan-Udall Foundation for the FDA, which focuses on drug safety research.

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Results: We found 21 gabapentin RCTs (74 reports, 6 IPD) and 7 quetiapine RCTs (50 reports, 1 IPD); most were reported in journal articles (18/21 [86%] and 6/7 [86%], respectively). When available, CSRs contained the most trial design and risk of bias information. CSRs and IPD contained the most results. For the outcome domains “pain intensity” (gabapentin) and “depression” (quetiapine), we found single trials with 68 and 98 different meta-analyzable results, respectively; by purposefully selecting one meta-analyzable result for each RCT, we could change the overall result for pain intensity from effective (standardized mean difference [SMD] = -0.45 ; 95% confidence interval [CI]: -0.63 to -0.27) to ineffective (SMD = -0.06 ; 95% CI: -0.24 to 0.12). We could change the effect for depression from a medium effect (SMD = -0.55 ; 95% CI: -0.85 to -0.25) to a small effect (SMD = -0.26 ; 95% CI: -0.41 to -0.1).

Conclusions: Disagreements across data sources affect the effect size, statistical significance, and interpretation of trials and meta-analyses. © 2017 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Keywords: Systematic reviews; Meta-analysis; Clinical trials; Reporting bias; Selective outcome reporting; Risk of bias assessment

1. Introduction

High-quality patient care depends on reliable, high-quality evidence of the effectiveness and safety of health care interventions. Although it is appropriate that we define high-quality evidence as coming from randomized clinical trials (RCTs) and systematic reviews of RCTs, the credibility of these sources may be challenged in two ways. First, although some RCTs are publicly reported (e.g., trial registrations, conference abstracts and other short reports, regulatory reviews, and journal articles), other RCTs are not reported publicly at all. Yet, nonpublic sources (e.g., trial protocols, clinical study reports [CSRs], and individual participant data [IPD]) often include important information about trial design, risk of bias, and results (Box 1). This means that decision makers do not have access to all information about many treatments. Second, even when RCT information is publicly available, information may be selectively reported, particularly results.

Multiple data sources create challenges for systematic reviewers and decision makers. On the one hand, current guidance is to examine all data sources to obtain complete information about each trial [6]. On the other hand, we know that information often differs among public sources to which systematic reviewers normally have access, and this challenge is compounded when one includes nonpublic sources [7,8]. When sources include different information about trial design, risk of bias, and results, the interpretation of individual RCTs may be affected, which in turn may impact the conclusions of systematic reviews and clinical practice guidelines.

Although previous studies have compared pairs of sources (e.g., data submitted to the Food and Drug Administration [FDA] compared with journal articles [9]), no study has compared all sources with one another. To our knowledge, no study has used all available sources to examine the extent to which systematic reviews might be sensitive to the sources that reviewers choose to use. As “open science” and data sharing continue to increase, systematic reviewers and decision makers need empirical information about the completeness and consistency of multiple data sources about RCTs so they can determine which sources to trust.

2. Objective

Our objective was to examine multiple data sources about RCTs assessing (1) gabapentin for neuropathic pain and (2) quetiapine for bipolar depression, to determine whether the information source could affect meta-analyses. We focused on these case studies because neuropathic pain and bipolar depression are prevalent conditions and because gabapentin and quetiapine are widely used drugs that are recommended in clinical guidelines for these conditions [10,11]. Additionally, a large number of data sources were available for both cases. These cases also represent two different ways that drugs are used in the United States: gabapentin is used off-label for many neuropathic pain conditions, while quetiapine is used on-label for bipolar depression.

3. Methods

Our goals were to use two case studies: (1) to determine whether multiple data sources for individual RCTs included the same or different information and (2) to assess whether disagreements across sources could affect the results of clinical trials and, thereby, the conclusions of systematic reviews and meta-analyses.

Our study was not a systematic review but was methodological research that studied the effect multiple sources of data can have on meta-analyses.

We published our protocol, including detailed methods [12], and have documented amendments (Supplement A at www.jclinepi.com).

3.1. Patient involvement

Two patient investigators were involved from the application to the funding agency forward; they are authors of this paper. Patient investigators contributed to selecting the eligibility criteria and outcomes and to interpreting the findings. To identify outcomes that matter to patients with neuropathic pain and bipolar depression, we also reviewed the website PatientsLikeMe (<http://www.patientslikeme.com>) and consulted public sources as described in the

What is new?**Key findings**

- Using two case studies, we found that multiple sources of clinical trial information often disagree, and differences in results in these sources can affect meta-analyses and the conclusions of systematic reviews.

What this adds to what was known?

- The wealth of information in nonpublic sources provides a thorough understanding of study design and risk of bias; it also allows “cherry picking” of results because myriad combinations of results allow multiple unique meta-analyses for any given outcome domain.

What is the implication and what should change now?

- “Open science” might increase the reliability of systematic reviews, but only if systematic reviewers develop methods for using additional data transparently and without bias. Our findings should inform future guidance for conducting systematic reviews (e.g., Cochrane, the U.S. National Academy of Medicine).

protocol [12,13]. Finally, we included one patient on our five-member Advisory Board.

3.2. Eligibility criteria

We considered trials eligible for our project if they would have been included in one of two possible systematic reviews: that is, RCTs were eligible if they compared (1) gabapentin or quetiapine with placebo or (2) gabapentin or quetiapine combined with another intervention with the other intervention alone. We excluded open-label and crossover trials to restrict our study to trials with comparable designs.

3.3. Identifying sources related to eligible trials

To identify journal articles and short reports (i.e., conference abstracts, summaries of journal articles, letters to the editor, posters, press releases, and reports in trade publications), we searched the Cochrane Central Register of Controlled Trials (CENTRAL), PubMed, Embase, LILACS, and CINAHL for each topic. We searched PsycInfo for quetiapine reports. We searched these databases through March 2, 2015 (gabapentin), and January 26, 2015 (quetiapine), without language restriction. For trial registrations, we searched the International Clinical Trials

Registry Platform Search Portal and ClinicalTrials.gov (October 10, 2014). We searched for medical and statistical reviews prepared by the FDA and made available on their website (Drugs@FDA). We checked reference lists of systematic reviews [14,15]. For gabapentin, we hand searched conference proceedings that were not included in our electronic searches, which we obtained from a coauthor (J.A.H.). For quetiapine, we did not conduct hand searches because of limitations in funding.

We identified nonpublic sources that were released during litigation for both drugs. For gabapentin, we included reports and IPD that were provided to a coauthor (K.D.) in her role as an expert witness. For quetiapine, we searched for reports online (<http://psychrights.org/>). We also wrote to gabapentin and quetiapine manufacturers (Pfizer and AstraZeneca) and requested nonpublic reports and IPD. Our correspondence with AstraZeneca has been published [16]. Our protocol provides additional information about searches for nonpublic sources [12].

We associated all identified sources with what appeared to be the appropriate trial(s). A single report could be associated with multiple RCTs; moreover, a single RCT could be associated with multiple reports.

3.4. Data extraction

From each eligible report, two authors independently extracted information about “trial design,” including participant characteristics, interventions, longest follow-up, trial funder, and author financial interests (<http://www.icmje.org/conflicts-of-interest>) and rated “risk of bias” items using the Cochrane risk of bias tool [17]. We extracted data using the Systematic Review Data Repository (<http://srdp.ahrq.gov/>) [18]. We reconciled differences through discussion.

We used the ClinicalTrials.gov framework to define and extract the five elements of each “outcome” [3,19]. That is, “outcomes” are often described by a “name” such as “pain intensity,” which is the “outcome domain.” The five elements that define an outcome are: (1) outcome domain (e.g., pain intensity); (2) measure (e.g., daily pain score, Short-Form McGill Pain Questionnaire); (3) metric (e.g., value at a time, change from baseline); (4) method of aggregation (e.g., continuous, categorical); and (5) time point of assessment (e.g., 8 weeks after starting treatment).

Reports often include both numerical data and other quantitative information such as “no statistically significant difference.” Here, we use the term “meta-analyzable result” to mean a numerical estimate of a treatment effect with sufficient information to calculate the between-group treatment effect (e.g., a point estimate and a measure of precision) [20]. We meta-analyzed only those outcomes for which we found sufficient data in a report to calculate the treatment effect. For example, we did not impute precision if it was not reported, and we did not meta-analyze data presented only graphically. The number of meta-

Box 1 Glossary of terms

Term	Definition used in our study
Clinical study report (CSR)	A special document type originating with drug and device manufacturers for submitting information to regulators (e.g., U.S. Food and Drug Administration, European Medicines Agency [EMA]). CSRs usually follow intentional guidelines for structure and content, and they often contain detailed summaries of trial design (including the trial protocol and statistical analysis plan) and results [1].
Individual patient data (IPD)	A record of all variables collected for each participant in a clinical trial (e.g., clinical characteristics, scores on tests and questionnaires), usually stored in a database file.
Outcome domain	The “name” describing a category of outcomes
Outcome	An event in a person, used to assess a treatment’s effect [2]. A defined outcome includes five elements [3]: (1) outcome domain, (2) specific measure, (3) specific metric, (4) method of aggregation, and (5) time point. For example “proportion of participants with 50% change from baseline to 8 weeks on the MADRS total score.”
Protocol	A document describing detailed plans for conducting a clinical trial, including the rationale, methods (e.g., for enrolling participants, delivering interventions, collecting outcomes, etc.), and ethical considerations [4]. Protocols are written before trials begin, and they may be updated during trials to reflect changes to study plans. For studies conducted by drug and device manufacturers, protocols may be included in CSRs.
Regulatory review	Documents written by agencies responsible for overseeing the manufacture and sale of medical products (e.g., FDA, EMA), which may contain both information supplied by manufacturers as well as information from other sources or analyses conducted by the agencies.
Report	A document describing the design, conduct, or results of a clinical trial. Reports include narrative information and are distinct from data sets, such as IPD.
Result	A numerical contrast between the test intervention and comparison groups (e.g., relative risk, mean difference). In this report, we describe only results that included sufficient information to calculate the between-group effect (e.g., a point estimate and a measure of precision).
Statistical analysis plan (SAP)	A document describing the planned methods for calculating the results of a clinical trial. Like protocols, SAPs may be written before trials begin and updated to reflect changes (e.g., methods for handling missing data might be revised in light of the observed data). For studies conducted by drug and device manufacturers, SAPs may be included in CSRs.
Trial registration	A record that includes structured information about an ongoing or completed trial [5]. A trial registration documents the existence of a clinical trial and provides an abridged summary of the trial design (e.g., planned sample size, primary and secondary outcomes). For completed trials, trial registrations may also include results.
Trial registry	A repository of trial registrations. An organized catalog of structured information about clinical trials.

Abbreviation: MADRS, Montgomery Åsberg Depression Rating Scale.

analyzable results in a trial report could be greater than the number of outcomes if there were multiple analyses (e.g., intention to treat, modified intention to treat) [20,21] or multiple treatment groups.

We extracted outcomes and meta-analyzable results for prespecified outcome domains. These included five outcome domains for gabapentin and eight outcome domains for quetiapine.

Gabapentin outcome domains:

- Mood
- Pain intensity
- Pain interference
- Quality of life
- Sleep disturbance

Quetiapine outcome domains:

- Anxiety
- Depression
- Functioning

- Psychiatric hospitalization
- Quality of life
- Sleep
- Suicide
- Weight gain

We extracted data at the time points closest to 8 (4–13), 18 (14–22), and 27 (23–31) weeks; that is, we did not extract every time point reported, and we did not consider data collected at baseline to be “meta-analyzable results.” While we extracted reports about more than one RCT, we did not extract results that combined data for multiple treatment groups or multiple RCTs. From IPD, we analyzed only the change from baseline for each measure.

3.5. Comparing trial design and risk of bias across multiple sources

We compared trial design described in each individual report (i.e., all sources except IPD). We identified two types

of differences: (1) difference in completeness, which we define as a difference occurring because information in one report was not available in other reports of the same trial (e.g., one report did not state the number of participants randomized, while another report did) and (2) substantive difference, which we define as contradictory information in two or more reports (e.g., two reports about the same trial reported that different numbers of participants were randomized).

We compared study quality among multiple reports of each trial.

To account for conflicting guidance about using short reports (e.g., conference abstracts) in systematic reviews [1,22], we performed a sensitivity analysis in which we compared only journal articles and nonpublic sources to determine if there were any disagreements in their descriptions of trial design and risk of bias.

3.6. Comparing trial results across multiple sources

When individual RCTs report multiple meta-analyzable results for a single outcome domain, systematic reviewers can conduct many meta-analyses (without changing which trials are included) by selecting different combinations of meta-analyzable results from the included trials.

To identify the potential impact of multiple results found in multiple data sources, we first counted the number of meta-analyses that could have been conducted for the pre-specified outcome domains using all data sources. That is, the number of possible meta-analyses is equal to the number of combinations of meta-analyzable results from each trial (i.e., the product of the number of meta-analyzable results).

To determine the range of meta-analytic effects that could be obtained using multiple sources and multiple results, we performed a resampling meta-analysis [23]. We conducted iterative meta-analyses as follows: (1) in each resampling iteration, we selected one meta-analyzable result from each RCT within the 8-week time window; (2) we combined the sampled meta-analyzable results for each iteration using a random-effects meta-analysis; (3) we iterated all possible meta-analyses using all meta-analyzable results from all sources or, when more than 10,000 combinations were possible, we randomly sampled 10,000 meta-analyses; and (4) we generated a distribution of the pooled estimates from meta-analyses. For each outcome, we conducted an overall analysis in which we included meta-analyzable results from all sources of all trials that reported the outcome, and we also conducted analyses limited to each source (e.g., journal articles, CSRs). For continuous outcomes, we calculated the standardized mean difference (SMD) for each domain, which allowed us to combine different measures; we conducted a sensitivity analysis using mean difference (MD) for the most common measure of pain and depression. For categorical

outcomes, we calculated the relative risk (RR) for each domain. For pain intensity and depression, we also calculated the risk difference for the most common measure in each case study. We used Stata 14 and R 3.3.1 to analyze the data.

To examine whether multiple data sources might influence the overall results of meta-analyses for a single outcome measure (which might be similar to an analysis that one might encounter in a typical systematic review), we conducted meta-analyses of the most common measures of pain and depression. For each case study, we compared (1) a meta-analysis using only data from journal articles with (2) a meta-analysis using data from all available sources in which we replaced the data from journal articles with data from nonpublic sources where possible.

An SMD or MD of zero means that the test treatment and the comparison have equivalent effects, and an SMD or MD less than zero indicates the degree to which the treatment is more efficacious than the comparison. An RR of one means that the test treatment and the comparison have equivalent effects, and an RR less than one indicates the degree to which the test intervention is more effective than the comparison.

4. Results

4.1. Most trials were associated with multiple sources

For gabapentin and quetiapine, we identified 21 and 7 eligible RCTs, respectively (Supplement B at www.jclinepi.com). We excluded some trials appearing in the most recent Cochrane review of gabapentin [14] for the following reasons: conditions were not definitively neuro-pathic (e.g., fibromyalgia), crossover design, and drug was gabapentin encarbil (a prodrug of gabapentin).

Of the included RCTs, 18/21 (86%) and 6/7 (85%), respectively, reported at least some information in a journal article by the time of our search, and 71% and 100% were associated with more than one report (Table 1). We identified 74 unique reports for eligible gabapentin trials, reported between 1997 and 2013, and 50 unique reports for eligible quetiapine RCTs, reported between 2003 and 2014 (Supplement C at www.jclinepi.com). Of the included reports, three and one, respectively, were not in English; co-authors who read these languages extracted relevant information. Gabapentin RCTs funded by the manufacturer were associated with many public reports. By contrast, RCTs not funded by the manufacturer were associated with only one or two reports (Table 1). All quetiapine RCTs were sponsored by the manufacturer.

For gabapentin, we located an FDA medical review and an FDA statistical review (101 and 23 pages, respectively) that described two included RCTs of gabapentin for post-herpetic neuralgia, an approved indication and type of neuropathic pain. FDA reviews did not include RCTs of

Table 1. Description of eligible trials and reports for gabapentin and quetiapine

	Gabapentin	Quetiapine
Number of trials	21	7
No. public reports/no. all reports ^a	68/74	46/50
Individual participant data (no. trials, % of total)	6 (29)	1 (14)
Trial characteristics (no. trials, % of total)		
Manufacturer funded ^b	14 (67)	7 (100)
≥3 groups ^c	11 (52)	4 (57)
Multicenter ^d	14 (67)	6 (86)
English language ^e	20 (95%)	7 (100)
Number of participants randomized (median, range) ^f	150 (26–452)	526 (100–802)
Sources of data for each trial (no. trials, % of all trials)		
Only public	15 (71)	3 (43)
Only nonpublic	1 (5)	0 (0)
Both public and nonpublic	5 (24)	4 (57)
Trials with each report type (no. trials, % of all trials)		
Journal article about 1 trial	17 (81)	6 (86)
Journal article about ≥2 trials	7 (33)	4 (57)
Short report: conference abstract	10 (48)	6 (86)
Short report: other	9 (43)	4 (57)
Trial registration	5 (24)	7 (100)
FDA report	2 (10)	0 (0)
CSR-Synopsis	0 (0)	2 (29)
CSR	6 (29)	2 (29)
Reports with results/all reports		
Journal article about 1 trial	18/20	10/10
Journal article about ≥2 trials	3/6	1/5
Short report: conference abstract	5/20	0/19
Short report: other	4/15	0/5
Trial registration	2/5	1/7
FDA report	2/2	Not applicable
CSR-Synopsis	Not applicable	2/2
CSR	6/6	2/2
Reports of manufacturer-funded trials ^a		
Manufacturer-funded trials (public reports per trial, SD)	7.4 (6.0)	10.3 (8.6)
Other trials (public reports per trial, SD) ^g	1.4 (0.5)	Not applicable

Public, Journal articles, conference abstracts, FDA reviews, trial registrations, other reports (letters to the editor, posters, press releases, reports in trade publications); Nonpublic, Clinical Study Reports (CSRs), CSR-Synopses.

^a Excludes individual participant data (IPD).

^b We examined all reports for each included trial and counted all trials with one or more reports stating that a manufacturer funded the trial as “manufacturer funded.” In addition to nine gabapentin trials funded by Pfizer, we identified four trials of an extended release formulation funded by Depomed and one trial funded by EpiCept (now Immune Pharmaceuticals), which manufactures a cream that was compared with gabapentin and placebo. All trials of quetiapine were sponsored by AstraZeneca.

^c Trials with one or more reports (public or nonpublic) describing at least three groups.

^d There was no information about the number of centers for four gabapentin trials.

^e Trials with one or more English language reports.

^f For trials reporting conflicting information, we selected the largest number of randomized participants for this analysis. The number of participants randomized was not reported for one quetiapine trial, which was not included in this analysis.

^g Includes one gabapentin trial reporting a nonmanufacturer funder and four gabapentin trials not reporting the funder in any source.

quetiapine for bipolar depression. For gabapentin and quetiapine, respectively, 5/21 (24%) and 7/7 (100%) trials were registered (all on ClinicalTrials.gov); for 2/5 (40%) and 1/7 (14%) registered trials, the investigators also deposited results on ClinicalTrials.gov. One registered trial of gabapentin was not reported in another public source at the time of our search.

We located nonpublic sources for 6/21 (29%) gabapentin and 4/7 (57%) quetiapine RCTs. Including appendices, the length of CSRs ranged from 1,315 to 8,027 pages.

As a component of the litigation discovery process [24], Pfizer provided information about all identified gabapentin RCTs conducted for any condition (including neuropathic

pain); only some of the 921 reports and 94 database files that Pfizer provided were relevant to our study. For eligible RCTs, we used CSRs (some of which contained protocols) and IPD. IPD were available as Microsoft Access databases without codebooks; Pfizer confirmed via email that codebooks could not be found. RCTs recorded different variables and used different coding schemes; furthermore, data structures differed across trials. For each RCT, we reconstituted the coding scheme using variable and table names, case report forms, trial protocols, statistical analysis plans, and other information in CSRs.

By searching online, we found CSRs for two quetiapine RCTs that had been released during litigation.

Only one of these CSRs included an appendix with eligible IPD, which we extracted using ABBYY FineReader software (version 11). For two other quetiapine RCTs, we identified CSR-Synopses that AstraZeneca supplied for a UK guideline about bipolar disorder [11]. In response to our request for information, AstraZeneca provided a list of reports about quetiapine; however, AstraZeneca refused our request for CSRs and IPD [16].

4.2. Trial design disagreed across sources

For individual RCTs, key information about trial design such as the number of treatment groups and the number of randomized participants sometimes disagreed across reports. Many short reports were inconsistent with other reports. Trial registrations were inconsistent with other sources, including 5/7 (71%) quetiapine registrations that did not describe the intervention groups completely

Table 2. Differences in trial design reported for trials with more than one report

Differences in study characteristics reported for trials with more than one report										
Gabapentin trials										
Trial	No. Reports	Participants				Intervention and Comparator(s)		Time (longest follow-up)	Trial funding	Financial Interests
		No. Rand.	Condition	Sex	Age	No. Groups	Treatment			
Arai 2010	2	X	O	O		X	X	O		
Backonja 1998	14	O	O	O	O	O	O	O	O	O
Caraceni 2004	3	O		O	O			O		
Hui 2010	2	X	O	O	O				X	X
Irving 2009	2									O
Study 945-224	6	O	O	O	O	O	O	O	O	O
Rice 2001	13	O	O	O	O	O	O	O	O	O
Rowbotham 1998	13	O	O	O	O	O	O	O	O	O
Sandercock 2012	3	O	O	O	O	X	X	O		
Sang 2013	17	O	O	O	O			O	O	O
Serpell 2002	10	X		O	O	X	X	O	O	O
Simpson 2001	2	O	O	O	O	X	X			
Tamez Pérez 2000	2	O	O	O	O		X			
Wallace 2010	16	X	O		O	X	X	O	O	O
Hahn 2004	1									
Milenkovic 2009	1									
Mishra 2012	1									
NCT00475904	1									
Study A945-1008	1									
Yildirim 2003	1									
Zepeda Vazquez 2001	1									
Not applicable (only one report)										

Quetiapine trials										
Trial	No. Reports	Participants				Intervention and Comparator(s)		Time (longest follow-up)	Trial funding	Financial Interests
		No. Rand.	Condition	Sex	Age	No. Groups ¹	Treatment			
Calabrese 2004	26	X	O	O	X	O	O	O		O
Gao 2014	2	X	O						O	
McElroy 2010	12	O	O	O	O	X	X			O
Suppes 2010	3	X	O	O	O	O	O			O
Thase 2006	17	X	O	O	O	X	X			O
Young 2008	14	O	O	O	O	X	X	O		O
LI 2014	2	O						O		O

X

Substantive difference across reports

O

Difference in completeness across reports

No difference across reports

We excluded IPD from this analysis and identified differences in completeness (blue) and substantive differences (red) across reports, including any case in which two or more reports contained different information. Each row represents a single trial, and each cell compares information available from all reports of that trial. No. reports, the number of reports related to each included trial; no. rand., number of participants randomized to all groups; condition, types of neuropathic pain or bipolar disorder (I or II) that were eligible for inclusion in trials of gabapentin and quetiapine, respectively; sex, proportion of male and female participants; age, mean age of participants; no. groups, number of treatment groups to which participants were randomized; treatment, drug and dose received by each group (we coded reports describing different numbers of groups as having a substantive difference in treatment); time (longest follow-up), longest planned follow-up time; trial funding, description of the funding for the trial; author financial interests, description of potential author interests rated using ICMJE categories (<http://www.icmje.org/conflicts-of-interest/>) and compared across reports as (1) any author declaring an interest, (2) all authors declaring no interest, or (3) authors not declaring if they have interests. ¹Five trial registrations described a single quetiapine group and do not state the dose; we rated the number of groups “unclear” (NCT00422214, NCT00206141, NCT00119652, NCT00083954, NCT00060489).

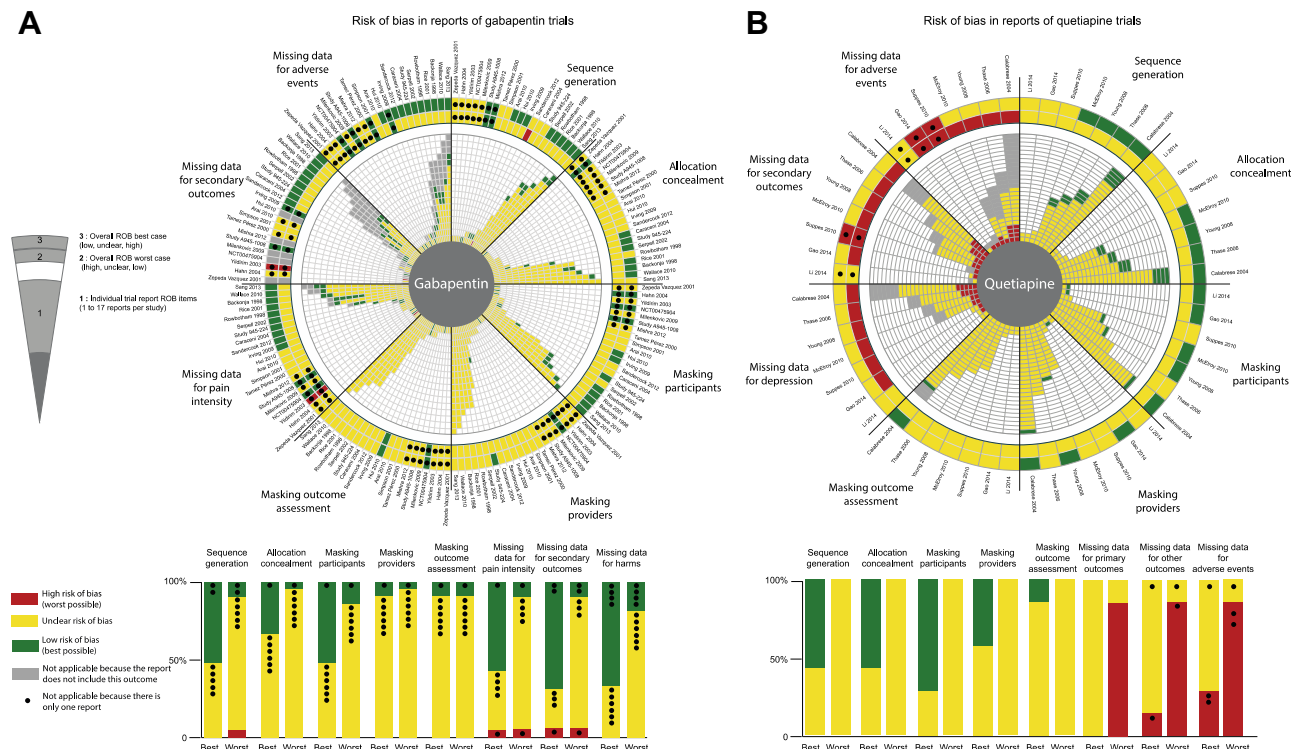


Fig. 1. Rating risk of bias using each individual report. There are eight sections in the circle, each representing one item that was assessed for risk of bias. Inner circle: Each “spoke” in a section represents a single trial, and each cell in a spoke represents a single report about the trial. Cells that are red, yellow, or green represent the rating of the risk of bias for that item using information from a single report. The sequence of trials is shown identically in each of the nine sections. Most cells are yellow, indicating that most risk of bias items could not be assessed in most reports. Gray cells appear only in sections about missing data; these reports did not mention outcomes to which the risk of bias item applies (e.g., adverse events). Circles indicate that a trial was associated with only one relevant report for a particular risk of bias item; and the number of trials with only one relevant report differs across the categories of “missing data” because reports included different outcomes. Outer circle: The two outer rings show the best and worst possible ratings of risk of bias. The inner ring is the worst risk of bias rating for each trial using information from a single report about the trial. The outer ring is the best risk of bias rating one could obtain using information from a single report for a trial. When the inner and outer ring differ, the risk of bias assessment for that item would be different depending on the source used. Barchart (bottom): The barchart summarizes the information in the outer circle. For each of the eight sections, the bars indicate the best and worst overall ratings of risk of bias using one report for each trial. Risk of bias for missing data was unclear in reports that did not describe the number of participants who withdrew from the trial: 16/74 (22%) reports about gabapentin trials included a flow diagram for an individual trial, and 29/74 (39%) stated the proportion of participants who withdrew before final assessment; 9/50 (18%) reports about quetiapine trials included a flow diagram, and 13/50 (26%) stated the proportion of participants who withdrew before final assessment. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

(Table 2). Although there were fewer disagreements in the sensitivity analysis in which we compared journal articles with nonpublic sources (Supplement D at www.jclinepi.com), we found that information in journal articles sometimes disagreed with other journal articles and with nonpublic sources about the same trial. The longest planned follow-up was often unclear, especially in short reports. There were few disagreements across sources in reported trial funder and author financial interests.

4.3. Risk of bias was unclear in most sources

We did not identify a source that reliably included all information needed to assess risk of bias. For RCTs with both journal articles and CSRs, CSRs included information about risk of bias that could not be found in any public report for 4/5 (gabapentin) and 1/2 (quetiapine) RCTs; journal articles provided

information that was not in the CSRs for 4/5 (gabapentin) and 0/2 (quetiapine) RCTs. Rather than use all sources to assess risk of bias when multiple sources are available for an RCT, systematic reviews could follow a hierarchy and extract information from a single source that contains the most complete information; unfortunately, in those cases for which both sources were available, a complete assessment sometimes required using both public and nonpublic sources.

As shown in Fig. 1 [25], most reports contained insufficient information for assessing risk of bias (see also the sensitivity analysis in Supplement E at www.jclinepi.com). Additional information from another source always changed a trial’s risk of bias rating of “unclear” to “low,” except for “missing data” items. In six quetiapine trials, missing data items changed from “unclear” to “high” when one trial report did not describe the amount of missing data, while another report about the same trial

Table 3. The number of possible meta-analyses for pain intensity (gabapentin) and depression (quetiapine)

Trial	Number of unique results for pain intensity (all sources)
Gabapentin trials, pain intensity outcome domain	
Arai 2010	No results
Backonja 1998	23
Caraceni 2004	No results
Hahn 2004	No results
Hui 2010	2
Irving 2009	38
Milenkovic 2009	No results
Mishra 2012	1
NCT00475904	1
Study 945-224	68
Study A945-1008	7
Rice 2001	57
Rowbotham 1998	24
Sandercock 2012	9
Sang 2013	6
Serpell 2002	17
Simpson 2001	No results
Tamez Perez 2000	No results
Wallace 2010	33
Yildirim 2003	1
Zepeda Vazquez 2001	NA
Total (number of combinations)	34,481,892,969,216
Trial	Number of unique results for depression (all sources)
Quetiapine trials, depression outcome domain	
Calabrese 2004	98
Gao 2014	11
Li 2014	No results
McElroy 2010	No results
Suppes 2010	No results
Thase 2006	96
Young 2008	9
Total (number of combinations)	931,392

For each trial, we recorded the number of different numerical results that could be included in a meta-analysis for the most common outcome domains (i.e., pain intensity for gabapentin, depression for quetiapine). For each trial, we counted results available from all data sources; when we found an identical result in more than one source for the same trial, we counted the result only once (i.e., this is the number of “unique results”). The number of possible meta-analyses for an outcome domain is equal to the number of combinations of results from all trials (i.e., the product calculated by multiplying the numbers of results from all trials).

stated that $\geq 25\%$ of participants were missing (i.e., these trials exceeded our threshold for the amount of missing data considered “high” risk of bias); all reports about the seventh trial were “unclear.” We found differences only in completeness across sources; there were no substantive differences in our assessments between sources.

4.4. Trials included multiple definitions of a few outcome domains

Two of our prespecified outcomes domains were not reported in any source: pain interference for gabapentin and psychiatric hospitalization for quetiapine.

Within each of the reported outcome domains, outcomes were defined in multiple ways, even holding the time point constant. For example, pain intensity and depression were assessed in all gabapentin and quetiapine trials. Because of variations in outcome measures, metrics, methods of aggregation, and because trialists used multiple methods of analysis [20], many meta-analyzable results could be associated with a single outcome domain. For the 8-week time window, for example, we obtained many as 68 different meta-analyzable results for “pain intensity” from a single trial of gabapentin and as many as 98 “depression” meta-analyzable results for a single trial of quetiapine (Table 3).

We were unable to conduct any meta-analysis for either case study that included all trials for two reasons. First, outcome definitions were not always comparable across trials, even within the same outcome domain. For example, pain intensity (gabapentin) was commonly assessed using the average daily pain score on an 11-point scale; however, some trials reported only categorical results (e.g., proportion with 50% reduction from the baseline), whereas other trials reported only continuous results (e.g., mean change from baseline). Second, the only available results for many outcomes included insufficient statistical information to combine them using meta-analysis. For example, some reports included mean values that were not accompanied by measures of precision.

4.5. Public sources did not include some meta-analyzable results that could be found in nonpublic sources

The majority of trials provided at least one meta-analyzable result in a public source (Table 1). However, nonpublic sources included outcomes and meta-analyzable results that did not appear in any public source (Supplement C at www.jclinepi.com). Most outcomes in nonpublic sources included meta-analyzable results (i.e., there was sufficient information to calculate the between-group treatment effect and its precision). Across all trials and all sources, we found 287 meta-analyzable results for pain intensity, but only 125 (44%) were reported in journal articles. We found 214 meta-analyzable results for depression, but only 38 (18%) were reported in journal articles [20].

4.6. Many meta-analyses were possible

As described above, multiple meta-analyses were possible when trials reported multiple outcome definitions and multiple methods of analysis. By calculating the number of possible combinations of meta-analyzable results, we found that we could have conducted more than 34 trillion unique meta-analyses of pain intensity using continuous results (SMD) from 14 trials, even after excluding identical results that appeared in more than one source. (Table 3 describes how we calculated the number of possible combinations.) Limiting our analysis to the most common measure, the average daily pain score on an 11-point scale, we could

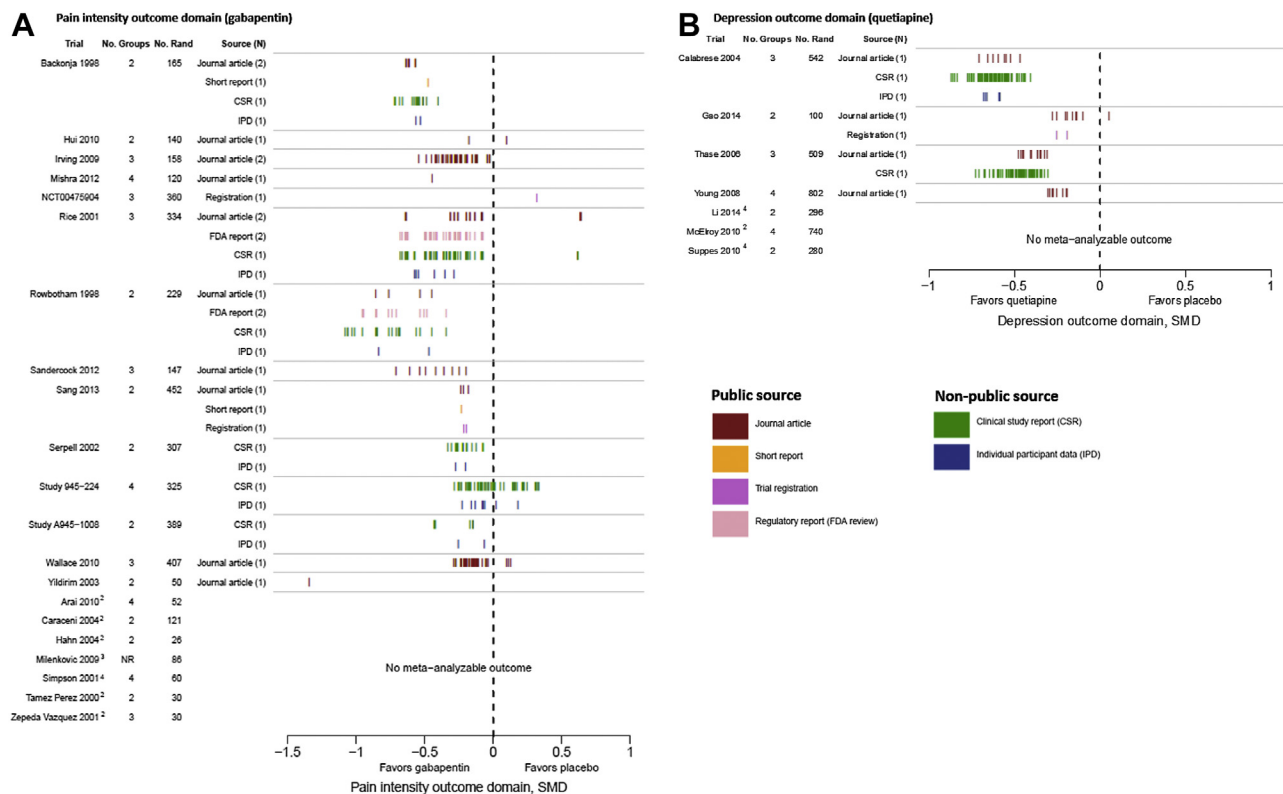


Fig. 2. Multiple results reported for the same outcome domain within reports and across multiple reports about trials of gabapentin and quetiapine. For each of the prespecified outcome domains, each line of a different color represents the SMD for a continuous outcome (1) for a single comparison (e.g., treatment vs. placebo) or (2) for two or more treatment groups (e.g., difference doses of gabapentin or quetiapine) vs. a comparator (which we calculated). For each trial, each row includes results for all sources of a given type (e.g., journal articles). Because some trials made more than one comparison and because some of the reported means were reported without defining each of the elements, the number of results reported for each trial sometimes exceeds the number of unique outcomes. The most extreme estimates from each trial are the effects that appear furthest to the left and furthest to the right. We did not include medians (57 gabapentin, 96 quetiapine). Results for other domains, and results for dichotomous pain and depression outcomes, are included in online supplement 7. ¹ Data come from a conference abstract. ² No result: no mean reported in any source for any outcome in this outcome domain. ³ No result: no mean reported within prespecified 8-week time window (4–13 weeks). ⁴ No result: no measure of precision reported with the reported mean. No. groups, number of treatment groups to which participants were randomized (not all of which were included in every report); no. rand, number of participants randomized (sometimes greater than the number included the analysis); source, type of sources available for each trial; N, for each type of source, the number of reports of that type; NR, information was not reported; SMD, standardized mean difference. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

have conducted over 352 million meta-analyses with 12 trials.

For quetiapine, we could have conducted 931,392 unique meta-analyses of depression using continuous results (SMD) from four trials (Table 3). Limiting our analysis to the most common measure, Montgomery Åsberg Depression Rating Scale, we could have conducted 4,797 meta-analyses with three trials.

In each case study, multiple meta-analyses were also possible for the other outcome domains (additional results in Supplement F at www.jclinepi.com).

4.7. Differences in results among multiple data sources affected results of meta-analyses

For any given trial, meta-analyzable results reported in CSRs, FDA reviews, and journal articles were broadly

consistent, as shown by the proximity of lines across sources in Fig. 2 (see also Supplement G at www.jclinepi.com). For gabapentin, trials included in FDA reviews found the largest effects (Fig. 3), and the results in the FDA reviews were consistent with other reports about the same trials (Fig. 2).

Meta-analyses were affected by multiple sources in several ways. First, using nonpublic sources increased the number of trials included in our analyses compared with using public sources alone. For example, using only journal articles, we were able to include 10 gabapentin and 4 quetiapine trials, respectively, in meta-analyses of the outcome domains pain intensity and depression. By comparison, we were able to include 14 and 4 trials using all sources. For the most common outcome measure in each case study, we were able to include 8 and 3 trials using journal articles compared with 12 and 3 trials using all sources.

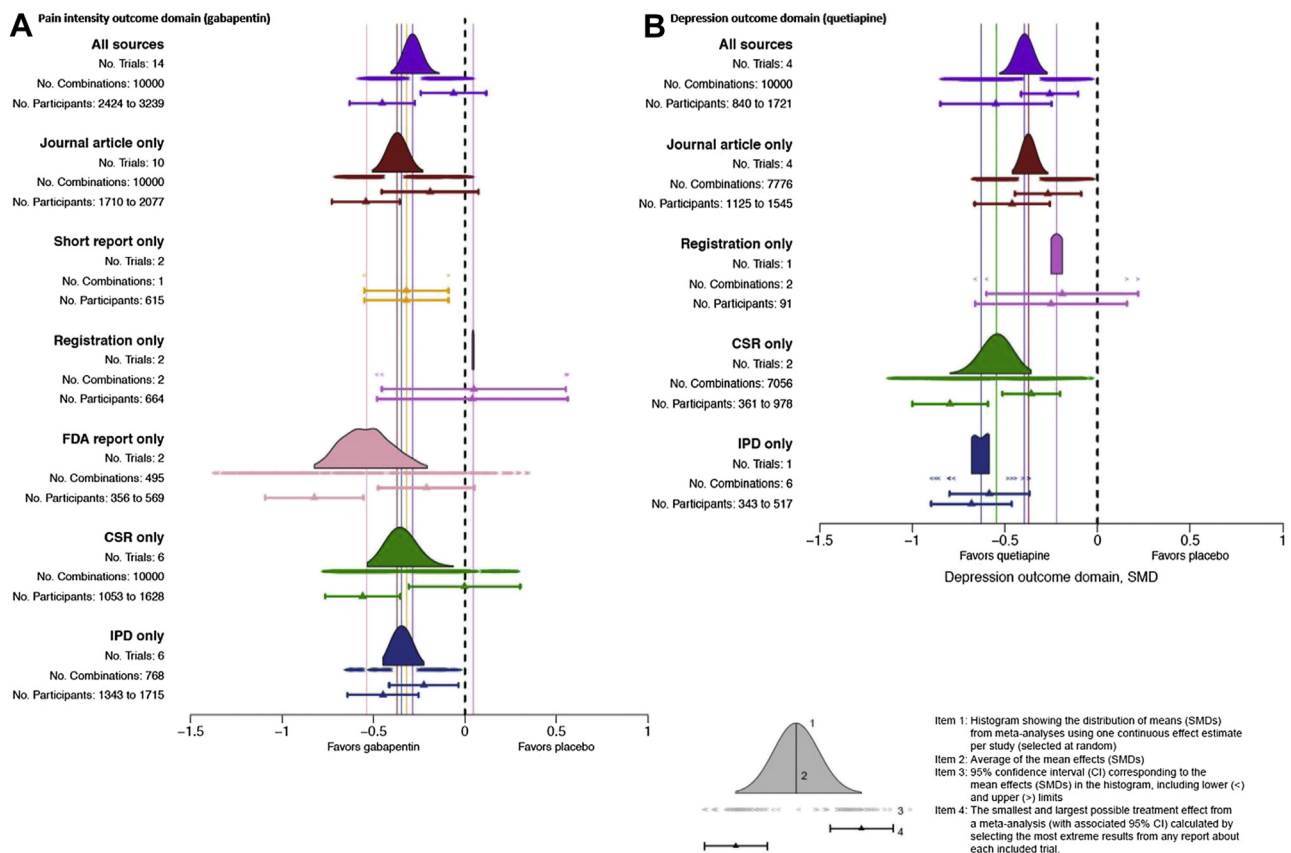


Fig. 3. Resampling meta-analysis of all possible meta-analyses that could have been conducted for the same outcome domain using multiple results and sources about trials of gabapentin and quetiapine. Using data from each source, distributions represent the means (SMDs) of all possible meta-analyses using one result per trial for continuous outcomes in one outcome domain. Because some trials made more than one comparison and because each outcome could be associated with multiple results (e.g., using different methods to handle missing data), some trials contributed multiple results for one outcome. For gabapentin and quetiapine, respectively, we did not include: 5 (24%) and 1 (14%) trials reporting no mean in any source (e.g., they reported only categorical data), 1 (5%) and 2 (28%) trials reporting no measure of precision with the reported mean, nor 1 (5%) and 0 (0%) trials reporting no analyzable data within the prespecified 8-week time window (4–13 weeks). Results for other domains and results for dichotomous pain and depression outcomes are included in online supplement 6. Compared with narrow histograms, wide histograms indicate that meta-analyses varied more in magnitude. When 95% confidence intervals for the treatment effects appear on both sides of the dashed line, it would be possible to conduct a meta-analysis with either a (1) statistically significant or (2) statistically nonsignificant result. The mean results (treatment effect) of the largest and smallest possible meta-analyses appear near the ends of the histograms (far left and the far right); when the 95% confidence intervals of these meta-analyses do not overlap, they are statistically different. No. trials, Number of trials included in each meta-analysis we performed; no. combinations, the number of meta-analyses represented in each plot. If multiple results were available within a trial (in one or more sources), we randomly select one result from each trial (within the 8-week time window); no. participants, the range in number of participants included in meta-analyses; CSR, Clinical Study Report; IPD, individual participant data.

Second, using resampling meta-analyses, we found that using different meta-analyzable results from the various sources could lead to different conclusions, especially if one focused on statistical significance. For example, using the various results reported for each trial (as shown in Fig. 2), one could purposefully select a favorable (far left in Fig. 2) or unfavorable (far right) result for each RCT. By cherry-picking results in this way, we could change the magnitude and statistical significance of the meta-analyses shown in Fig. 3. For pain intensity, the largest possible estimate (far left in Fig. 3) was consistent with a medium effect for gabapentin compared with placebo and was statistically significant (SMD = -0.45 ; 95% confidence interval [CI]: -0.63 to -0.27). The largest possible estimate did not overlap with the smallest possible estimate

(far right); the latter was consistent with no effect and was not statistically significant (SMD = -0.06 ; 95% CI: -0.24 to 0.12). For depression, the largest possible estimate was consistent with a medium effect for quetiapine compared with placebo (SMD = -0.55 ; 95% CI: -0.85 to -0.25) and overlapped with the smallest possible estimate, which was consistent with a small effect (SMD = -0.26 ; 95% CI: -0.41 to -0.1); both estimates for depression were statistically significant. Purposefully selecting the largest or smallest estimates always changed the magnitude of effect sizes for both continuous and categorical outcomes, and it changed the statistical significance of results for 3/4 (75%) and 2/7 (29%) outcome domains reported in trials of gabapentin and quetiapine, respectively (Supplement F at www.jclinepi.com).

Table 4. Differences between meta-analyses of the most common outcome measures with and without nonpublic sources

Trial	Estimate including only journal articles (SMD, 95% CI)	Estimate including all sources (SMD, 95% CI)
Gabapentin trials, average daily pain score on an 11-point scale		
Backonja 1998	−0.57 (−0.88 to −0.25)	−0.53 (−0.85 to −0.22)
Irving 2009	−0.37 (−0.71 to −0.04)	−0.37 (−0.71 to −0.04)
Mishra 2012	−0.44 (−0.95 to 0.06)	−0.44 (−0.95 to 0.06)
NCT00475904	No reports	0.32 (0.04 to 0.60)
Rice 2001	−0.64 (−0.92 to −0.36)	−0.56 (−0.80 to −0.33)
Rowbotham 1998	−0.86 (−1.13 to −0.59)	−0.83 (−1.11 to −0.56)
Sandercock 2012	−0.49 (−0.84 to −0.15)	−0.49 (−0.84 to −0.15)
Sang 2013	−0.22 (−0.40 to −0.03)	−0.22 (−0.40 to −0.03)
Serpell 2002	−0.86 (−1.13 to −0.59)	−0.20 (−0.43 to 0.03)
Study 945-224	No reports	−0.06 (−0.32 to 0.19)
Study A945-1008	No reports	−0.26 (−0.46 to −0.06)
Wallace 2010	−0.20 (−0.40 to 0.01)	−0.20 (−0.40 to 0.01)
Total	−0.46 (−0.64 to −0.28)	−0.31 (−0.47 to −0.15)
Quetiapine trials, Montgomery Åsberg Depression Rating Scale		
Calabrese 2004	−0.63 (−0.84 to −0.41)	−0.59 (−0.78 to −0.40)
Thase 2006	−0.37 (−0.56 to −0.17)	−0.46 (−0.66 to −0.27)
Young 2008	−0.27 (−0.46 to −0.08)	−0.27 (−0.46 to −0.08)
Total	−0.42 (−0.62 to −0.22)	−0.44 (−0.62 to −0.26)

Abbreviation: CI, confidence interval.

Results for all trials that reported the most common outcome measure of pain intensity (the average daily pain score on an 11-point scale) or depression (Montgomery Åsberg Depression Rating Scale) within the 8-week time window in any source. For each included trial, we calculated the standardized mean difference (SMD) and 95% CI for the intervention compared with placebo. We combined the results (“Total”) by conducting a meta-analysis using random effects. We first estimated the effect using only journal articles; we then estimated the effect using all sources. For the analysis using all sources, we replaced estimates from journal articles with estimates from nonpublic sources where both were available. For example, we used a prespecified hierarchy to select outcome definitions and methods of analysis; we prioritized the metric “change from baseline” over “value at a time,” and we prioritized “intention to treat analyses” over “per protocol” other methods of analysis [12].

Finally, comparing the overall results for a single outcome measure, we found that by using all sources and by replacing public sources with nonpublic sources, we reduced the magnitude of the effect for the most common measure for gabapentin but not quetiapine (Table 4). The effect of gabapentin on the average daily pain score using only journal articles (SMD = −0.44; 95% CI: −0.63 to −0.29) was greater than the effect using all sources (SMD = −0.31; 95% CI: −0.47 to −0.16). The effect of quetiapine on the Montgomery Åsberg Depression Rating Scale was essentially the same whether we used only journal articles (SMD = −0.42; 95% CI: −0.62 to −0.22) or all sources (SMD = −0.44; 95% CI: −0.62 to −0.26).

5. Discussion

In our two case studies, not all trials were reported publicly, and multiple data sources were available for most trials. Comparing all data sources, we found that information about trial design (e.g., number of treatment groups) and risk of bias often differed in completeness across reports, and reports sometimes disagreed about trial design. Detail-rich nonpublic sources included more information than public sources, and multiple sources were often necessary to obtain all available information about risk of bias. Following current best practices for performing systematic reviews of

RCTs was both resource-intensive and confusing because conflicting information could be difficult to disentangle.

Differences in results across sources suggested selective outcome reporting. For example, no source presented meta-analyzable results for pain intensity and depression for several trials of gabapentin and quetiapine, respectively. Many meta-analyzable results were available only in nonpublic sources and when these results were added to a meta-analysis of data from similar trials, they could change the summary results. Although results for a single outcome and method of analysis in a single trial were usually consistent across multiple sources, results available only in nonpublic sources were often less favorable to the test interventions. A previous study about gabapentin found evidence of a deliberate strategy to distort the scientific literature [26].

While many studies have shown that selective reporting is a problem, only a few studies have explored the potential for systematic reviewers to select outcomes for meta-analysis, [23,27,28] and none has used both public and nonpublic sources. By cherry-picking results for meta-analyses, we found that systematic reviewers could change the results of meta-analyses and the conclusions of systematic reviews. In the case of gabapentin, the two most extreme analyses made the drug appear either highly effective or completely ineffective for treating neuropathic pain. On the other hand, if one focused only on statistical

Box 2 Recommendations to stakeholders based on our findings

Patients and clinicians

- Be aware that multiple publications may be related to a single trial
- Do not rely on conference abstracts alone for information about intervention safety and effectiveness
- Consider the trustworthiness of trial data and meta-analyses when making decisions

Trialists

- Anticipate data sharing at each stage of the trial (e.g., registration, publishing the protocol, data collection, and documentation)
- Share individual patient data (IPD) and the meta-data needed to understand and analyze IPD
- Define outcomes completely, including all five elements (i.e., outcome domain, specific measure, specific metric, method of aggregation, and time point)
- Report results including sufficient information for meta-analyses
- Adhere to reporting guidelines (e.g., for protocols and trial reports)

Systematic reviewers

- Develop a protocol that reflects expectations about multiple sources of trial information (e.g., short reports, clinical study reports [CSRs])
- Determine whether nonpublic sources will be used for studies that are also reported in journal articles
- If seeking nonpublic sources, make necessary enquiries early
- Anticipate handling multiple data sources for individual trials and allow time for assessing risk of bias and for extracting data
- Prespecify outcomes, including all five elements
- Plan for selecting among related outcomes within trials and for combining different outcomes across trials

Other decision makers (e.g., journal editors, policymakers, funders)

- For new studies, require complete and prospective trial registration and require sharing of data sets (including meta-data), codebooks, trial protocols, statistical analysis plans, and statistical code
- For completed (“legacy”) trials, require sharing of at least CSRs and summary (i.e., aggregate) data

significance and ignored the magnitude of effect estimates, the conclusion that quetiapine reduces bipolar depression at 8 weeks was not affected by data source. We observed similar variations in overall results for other outcome domains, for aggregation of both continuous and categorical data (see [Supplements F and G](#) at www.jclinepi.com). These results suggest that guidance for systematic reviews should be updated to help reviewers use multiple sources efficiently and to minimize biased outcome selection in meta-analyses (see [Box 2](#)) [20,29].

Using all sources, we did not identify a single source or combination of sources that always provided the information needed to conduct systematic reviews. Among public sources, conference abstracts were often unclear or misleading when used on their own (e.g., because they misstated the number of participants). Journal articles were good sources of information when compared with nonpublic sources, although they did not include complete information about trial design, risk of bias, or results. For a single trial, journal articles were sometimes inconsistent with other journal articles and with nonpublic sources. Although FDA reviews provided detailed results for on-label uses of gabapentin, they were not available for most trials in either case study. Checking ClinicalTrials.gov was useful for identifying trials, but records were often incomplete and contained information about trial design that contradicted information in nonpublic sources.

Our findings have important implications for data sharing and for the use of multiple data sources in systematic reviews (see [Box 2](#)) [29]. Using nonpublic sources usually improved our understanding of study design and risk of bias and changed

our estimates of clinical effectiveness. Although IPD has been described as the “gold standard” for systematic reviews [17], aggregate data in CSRs (which contained the most information about risk of bias) would be sufficient for many purposes. IPD may be necessary when researchers want to explore new questions, properly account for multiple outcomes, conduct more extensive reviews of adverse event data, and use modern techniques to adjust for missing data [30–32].

Our findings may be generalizable to other treatments and conditions. The nonpublic sources we obtained were available because of litigation, an approach that has been used in previous studies to compare data sources [33]. Our findings are broadly consistent with previous studies comparing various sources with trial registrations [34–36], conference abstracts [37,38], regulatory reviews [39,40], journal articles [41], protocols [42–44], CSRs [24,45], and IPD [46–48].

Our findings have important implications for stakeholders seeking trial information ([Box 2](#)). Although we had nonpublic sources for only 29% of gabapentin RCTs, the addition of nonpublic sources changed our conclusions, demonstrating that increased research transparency may be useful even if multiple sources are not available for all trials in a systematic review. This is an especially important finding for a drug that is recommended in clinical guidelines [10] and under consideration for the World Health Organization list of essential medicines [10,49]. For quetiapine, the addition of nonpublic sources did not lead to a meaningfully different conclusion about whether it has some effect on depression compared with placebo. However, quetiapine is recommended in clinical guidelines

that also consider other interventions [11], and further research would be needed to determine whether the differences we identified would be sufficient to change recommendations about its use in practice (e.g., to determine whether it would appear to be less cost-effective compared with other interventions).

Our findings demonstrate the value of open science [50], including data sharing [51] and prespecifying analysis plans [52]. Providing access to more complete trial information can increase the reliability of systematic reviews and improve the accuracy of treatment effect estimates. Although many discussions have focused on recent trials (e.g., since 2007 [53]), our findings underscore the importance of sharing completed “legacy” trials for widely used drugs [54]. We found it particularly difficult to access this information, and we conclude that the scientific community should develop systems for archiving and sharing data and associated meta-data for legacy trials before they are lost forever. For both new trials and legacy trials, standards are needed to make evidence not only open but useful (e.g., a “code of conduct”) [55].

Data sharing alone will not solve all the problems we identified. First, trialists must present data consistently across sources. Second, trialists and systematic reviewers must prespecify all outcomes (i.e., using all five elements) [3,19] and methods of analysis (e.g., by sharing a “statistical analysis plan” before analyzing the data) [19,20] and report changes to their plans [27,56,57]; otherwise, results from RCTs will be open to “cherry picking” by both trialists and systematic reviewers [20]. Third, systematic reviewers must decide whether and how they will use all available data sources. Although data sharing [54,58,59], trial registration, [60–62] and reanalysis of IPD [63] have been discussed for decades, our findings indicate that further discussions are needed to coordinate the many steps involved.

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Authors’ contributions: The study design was first described in the application to the Patient-Centered Outcomes Research Institute (PCORI) in 2013. K.D. was the principal investigator and worked with T.L., S.V., and P.D. to design the study, write the application, and obtain the funding. E.M.-W. drafted the protocol with

contributions from other authors. T.C., K.D., N.F., G.G., J.A.H., J.H., D.P., T.L., E.M.-W., J.L.P., and E.T. contributed to developing the data extraction forms. L.R. and C.T. designed and ran the electronic searches. K.D., N.F., G.G., J.H., E.M.-W., T.L., and S.V. screened studies for inclusion. L.B., J.K.C., J.E., N.F., G.G., J.H., D.P., E.M.-W., and C.S.-C. extracted data. J.K.C., N.F., H.H., N.G., and E.M.-W. managed the data. J.K.C., N.F., N.G., and H.H. analyzed data. K.D., J.E., N.F., H.H., T.L., E.M.-W., and E.A.S. contributed to interpretation and data presentation. E.M.-W. wrote the first draft, with E.M.-W., T.L. and K.D. providing subsequent revisions. All authors reviewed, provided critical revisions, and approved the manuscript for publication. E.M.-W. is the guarantor. All authors, external and internal, had full access to all of the data (including statistical reports and tables) in the study and can take responsibility for the integrity of the data and the accuracy of the data analysis.

Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jclinepi.2017.07.014>.

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